

NOTES

Probability and Distributions



Probability and Distributions

Introduction To Probability

What is Probability?

- Probability measures how likely an event is to occur.
- The value of probability ranges from 0 to 1 where:
 - 0 means the event will never happen (impossible).
 - 1 means the event will definitely happen (certain).

Key Points About Probability

Range of Probability:

- **Probability values always lie between 0 and 1**, inclusive. and this can also be expressed as percentages between **0% and 100%**.

Examples:

- A probability of **0.75** is the same as saying there's a **75% chance** of the event happening.
- A probability of **0** means the event will **not occur** under any circumstances.

Usefulness of Probability:

- Probability helps us **predict outcomes** in situations where there is uncertainty, such as:
 - Rolling dice
 - Flipping coins
 - Drawing cards

Basic Probability Formula

The formula for calculating the probability of an event is:

$$P(\text{Event}) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$$

Components:

- **Favorable Outcomes:** The specific outcomes we are interested in.
- **Total Possible Outcomes:** All possible outcomes that can occur in the situation.

Example : What is the probability of rolling a 3 on a fair six-sided die?

When rolling a fair six-sided die, the total possible outcomes are the six numbers on the die that is 1,2,3,4,5,6. Therefore, there are 6 possible outcomes in total. The event we are interested in is rolling a 3, and there is only one favorable outcome for this event: rolling a 3. To find the probability of rolling a 3, we divide the number of favorable outcomes (which is 1) by the total number of possible outcomes (which is 6). So, the probability of rolling a 3 is:

$$P(3) = \frac{1}{6}$$

Example : What is the probability of getting Heads when you flip a coin?

When flipping a coin, there are two possible outcomes: Heads or Tails. So, the total number of possible outcomes is 2. The event we are interested in is getting Heads, and there is only one favorable outcome: the coin landing on Heads. To calculate the probability, we divide the number of favorable outcomes (1) by the total number of possible outcomes (2). Therefore, the probability of getting Heads is:

$$P(\text{Heads}) = \frac{1}{2}$$

Final Answer:

The probability of getting Heads when you flip a coin is $\frac{1}{2}$ or 0.5 (which means 50%).

Conditional Probability

What is Conditional Probability?

Conditional probability refers to the probability of an event happening given that another event has already occurred. In other words, it is the chance of

something happening, but this time considering some known information about another event that has already taken place.

For example, imagine you're at school, and it's cloudy outside. Normally, there's a certain chance of rain, but because it's already cloudy, the chance of rain is higher. The probability of rain now depends on the fact that it's cloudy. This is a typical case of conditional probability, where the occurrence of one event (cloudy weather) affects the probability of another event (rain).

In terms of notation, conditional probability is written as $P(A|B)$, which means "the probability of event A happening, given that event B has already happened." In this example, A could be "it rains" and B could be "it's cloudy." So, $P(\text{Rain} | \text{Cloudy})$ represents the probability of rain, given that it is cloudy.

How to Calculate Conditional Probability

The formula is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A \cap B)$: Probability that both A and B happen.
- $P(B)$: Probability that B happens.

Understanding Conditional Probability with Two Approaches (Step-by-Step Solution)

We'll solve the problem of picking marbles using two approaches:

- **Direct reasoning** based on the sequence of events.
- **Using the formula** for conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

By combining both approaches, you'll clearly understand the logic and how the formula works.

The Problem:

You have a bag containing:

- 5 red marbles
- 5 blue marbles

You randomly pick two marbles, one after the other, without putting the first marble back.

Question: What is the probability of picking a blue marble on the second draw, given that the first marble picked is blue?

First Approach: Direct Reasoning

We solve this by reasoning step-by-step:

- **First event (picking a blue marble):**
 - There are 5 blue marbles out of 10 total marbles.
 - Probability of the first event: $5/10 = 0.5$
- **Second event (picking another blue marble, given the first was blue):**
 - After the first blue marble is taken, only 4 blue marbles remain.
 - The total number of marbles left is now 9.
 - Probability of the second event: $4/9$
- **Conditional Probability:**
 - Since the first event (picking the first blue marble) has already happened, the conditional probability of the second event is:

$$P(A|B) = \frac{4}{9}.$$

This direct reasoning approach gives us the answer directly.

Second Approach- Using Formula

We now use the **conditional probability formula** to solve the same problem:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Step 1: Find $P(A \cap B)$

- $P(A \cap B)$: Probability that both the first and second marbles are blue.
- To calculate this, we multiply the probabilities of the two events happening together:
 1. **First event (picking a blue marble):** Probability = $\frac{5}{10} = 0.5$.
 2. **Second event (picking another blue marble after the first is blue):** Probability = $\frac{4}{9}$.

So,

$$P(A \cap B) = P(\text{First Blue}) \cdot P(\text{Second Blue} \mid \text{First Blue}) = \frac{5}{10} \cdot \frac{4}{9} = \frac{20}{90} = \frac{2}{9}.$$

Step 2: Find $P(B)$

- $P(B)$: Probability of the first event (picking a blue marble).
 - We already know $P(B) = \frac{5}{10} = 0.5$.

Step 3: Calculate $P(A|B)$

Now substitute the values into the formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{9}}{0.5}.$$

Simplify:

1. Divide $\frac{2}{9}$ by 0.5 (which is the same as multiplying by 2):

$$P(A|B) = \frac{2}{9} \cdot 2 = \frac{4}{9}.$$

What is Bayes' Theorem?

- Bayes' Theorem helps us calculate the probability of something happening based on new information, essentially updating our belief or understanding after receiving new evidence.

Basic Concept:

- Imagine you have some prior knowledge or belief about something, known as the "prior probability." Then, you receive new evidence, and Bayes' Theorem helps you update your belief based on that evidence.

Formula for Bayes' Theorem:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- $P(A|B)$: The probability of **A** happening given that **B** has happened.
- $P(B|A)$: The probability of **B** happening given that **A** has happened.
- $P(A)$: The initial or prior probability of **A** happening.
- $P(B)$: The total probability of **B** happening.

Let's Understand with a Simple Example:

- Imagine you have two bags of marbles: Bag 1 contains 3 red marbles and 2 blue marbles, while Bag 2 contains 4 red marbles and 6 blue marbles.
- You randomly pick one of the bags (without knowing which one you picked) and then draw a blue marble.
- Now, you want to know the probability that you picked Bag 2, given that you drew a blue marble. This is where Bayes' Theorem comes into play, as it helps you update the probability based on the new evidence—picking a blue marble.

Step-by-Step Explanation Using Bayes' Theorem

Step 1: What Do We Know?

- There's an equal chance (50%) of picking either Bag 1 or Bag 2. So,

$$P(\text{Bag 1}) = P(\text{Bag 2}) = 0.5$$

- If you picked Bag 1, the chance of drawing a blue marble is $P(\text{Blue} | \text{Bag 1}) = \frac{2}{5}$ because there are 2 blue marbles out of 5 total marbles.
- If you picked Bag 2, the chance of drawing a blue marble is $P(\text{Blue} | \text{Bag 2}) = \frac{6}{10} = 0.6$ because there are 6 blue marbles out of 10.

Step 2: Use Bayes' Theorem

We want to find:

$$P(\text{Bag 2} | \text{Blue}) = \frac{P(\text{Blue} | \text{Bag 2}) \cdot P(\text{Bag 2})}{P(\text{Blue})}$$

What We Know:

- **Bag 1:** 2 blue marbles, 3 red marbles (total of 5 marbles)
 - Probability of picking a blue marble from **Bag 1** = $\frac{2}{5}$
- **Bag 2:** 6 blue marbles, 4 red marbles (total of 10 marbles)
 - Probability of picking a blue marble from **Bag 2** = $\frac{6}{10} = 0.6$
- There's a 50% chance of picking **either** Bag 1 or Bag 2 (since you choose randomly).

Formula for Total Probability (P(Blue)):

To find the **total probability of picking a blue marble**, we need to add up the probabilities from both bags. The rule for this is:

$$P(\text{Blue}) = P(\text{Blue} \mid \text{Bag 1}) \cdot P(\text{Bag 1}) + P(\text{Blue} \mid \text{Bag 2}) \cdot P(\text{Bag 2})$$

- $P(\text{Blue} \mid \text{Bag 1})$ is the probability of picking a blue marble from **Bag 1**.
- $P(\text{Bag 1})$ is the probability of choosing **Bag 1** (which is 0.5 because it's a 50% chance).
- $P(\text{Blue} \mid \text{Bag 2})$ is the probability of picking a blue marble from **Bag 2**.
- $P(\text{Bag 2})$ is the probability of choosing **Bag 2** (also 0.5).

Substituting the values:

$$P(\text{Blue}) = \left(\frac{2}{5} \cdot 0.5\right) + (0.6 \cdot 0.5)$$

$$P(\text{Blue}) = 0.2 + 0.3 = 0.5$$

Now, using Bayes' Theorem:

$$P(\text{Bag 2} \mid \text{Blue}) = \frac{0.6 \cdot 0.5}{0.5} = \frac{0.3}{0.5} = 0.6$$

Final Answer:

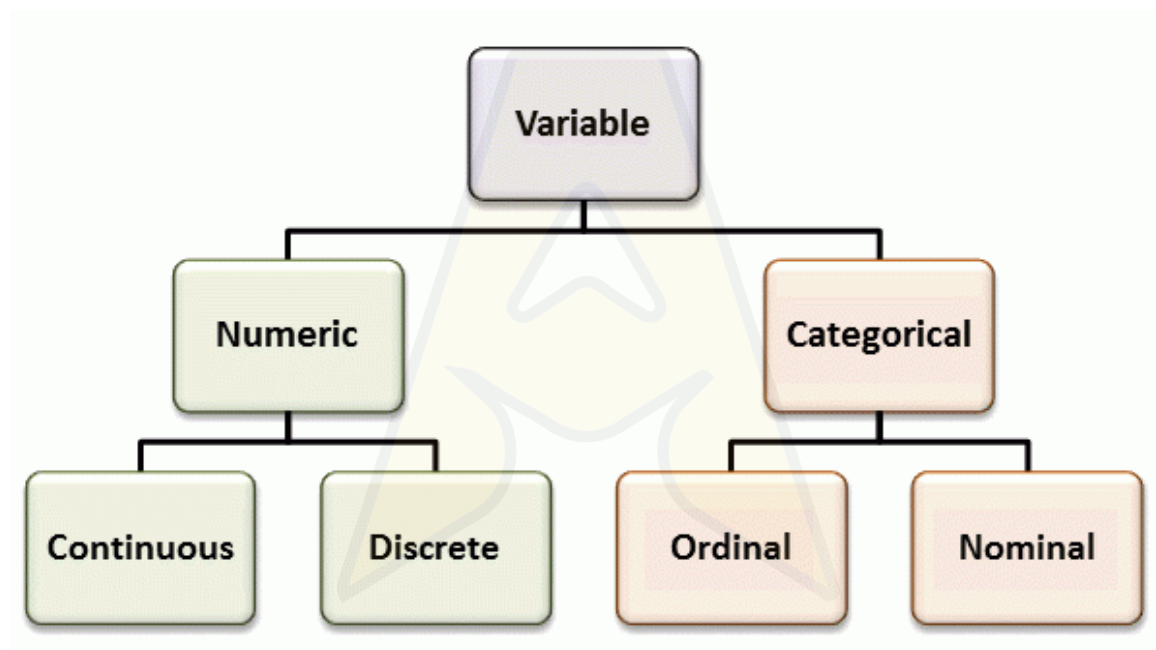
Given that you picked a blue marble, there's a **60% chance** you picked **Bag 2**.

In statistics, a variable is a property or characteristic that can change or vary and is something we can measure or observe. Variables can take on different values depending on what we are studying. For example, the height of students in a classroom, the number of apples in a basket, or the favorite colors of people in a group are all variables. In simple terms, a variable is something that changes and is measured or tracked in a study.

Types Of Variables:

Variables in statistics are typically divided into two main types:

- **Quantitative Variables** (also called Numerical Variables)
- **Qualitative Variables** (also called Categorical Variables)



Let's look at these two types in more detail, with easy-to-understand examples.

1. **Quantitative Variables** (Numerical Variables)

- Quantitative variables represent numbers or quantities that can be counted or measured, answering questions like "How many?" or "How much?" For example, the number of students in a class or the amount of water in a bottle are both examples of quantitative variables.

Types of Quantitative Variables:

- **Discrete Variables**

Discrete variables are those that can only take whole numbers or countable values, representing things we can count but not fractions or decimals. For example, the number of students in a classroom is a

discrete variable—you can have 10 students or 20 students, but not 15.5 students.

- **Continuous Variables**

Continuous variables represent quantities that can take any value within a range and are measurable, allowing for decimals or fractions. For example, the height of a person is a continuous variable—it could be 5.6 feet, 5.62 feet, or 5.623 feet.

2. **Qualitative Variables** (Categorical Variables)

- Qualitative variables represent categories or labels that describe something and cannot be measured in terms of numbers. They answer questions like "What type?" or "Which one?" Examples include the color of a shirt, the gender of a person, or the country someone is from.

Types of Qualitative Variables:

- **Nominal Variables** are categories with no natural order or ranking. These variables are simply labels without any hierarchy.

Example:

- If we ask people about their favorite fruit:
- Person A likes Mango.
- Person B likes Banana.
- Person C likes Apple.

The variable "favorite fruit" is **nominal** because the fruits are just categories, with no ranking.

- **Ordinal Variables**

- Ordinal variables are also categories, but in this case, there is a natural order or ranking to the categories.

- Example: - Movie ratings like "Poor," "Good," and "Excellent" are ordered: - Poor < Good < Excellent - However, they don't specify by how much one is better than the other.

Why Are Variables Important in Statistics?

- **Understanding Data:** Identifying variable types guides the choice of analysis tools:
 - **Quantitative data:** Use histograms or scatter plots (e.g., age, weight).
 - **Qualitative data:** Use bar charts or pie charts (e.g., favorite colors, movie ratings).
- **Making Decisions:** Understanding variables aids predictions and decisions. For example, businesses analyze sales data (discrete) to optimize inventory

levels.

Real-Life Examples of Variables in Health, Education, and Data Science

- **Health:**

Height and weight, as continuous variables, are commonly used to assess health metrics such as Body Mass Index (BMI). Similarly, health data like blood pressure and age are often incorporated into predictive models to evaluate risks for conditions like heart disease or diabetes.

- **Data Science:**

- Quantitative variables like income, age, and ratings are often used to make predictions, while qualitative variables like feedback and categories help segment data for analysis.
- For example, in movie recommendations, variables such as genre (nominal), ratings (ordinal), and views (discrete) are used to personalize suggestions.

In all fields, understanding variables is key to accurate analysis and decision-making.

2. Random Variables

- A random variable is a numerical representation of the possible outcomes of a random event. It helps translate the results of random occurrences into measurable and analyzable data.

For Example:

- If you flip a coin, you might call **Heads** "1" and **Tails** "0."
- If you roll a die, the number that shows up (like **1**, **2**, or **6**) is a random variable.

In Simple Terms:

A random variable is just a **number that represents the result** of something uncertain.

Two Types of Random Variables

There are two main types of random variables, depending on the kind of outcomes they represent:

1. **Discrete Random Variables**

2. Continuous Random Variables

Let's understand both step by step.

1. Discrete Random Variable

A **discrete random variable** can only take certain specific values. These values are **countable**, and there are no "in-between" possibilities.

Examples:

Number of Heads in 3 Coin Tosses:

Imagine tossing a coin three times, where the random variable X represents the number of heads obtained. The possible outcomes of the coin tosses are: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT.

Count the heads in each outcome:

- ($X = 0$): TTT (no heads)
- ($X = 1$): HTT, THT, TTH (1 head in each case)
- ($X = 2$): HHT, HTH, THH (2 heads in each case)
- ($X = 3$): HHH (all heads)

Here, the random variable (X) can only take the values **0, 1, 2, or 3**. These are countable, so this is a discrete random variable.

Rolling a Die:

Let Y represent the number that appears when rolling a six-sided die. The possible values of Y are 1, 2, 3, 4, 5, or 6, corresponding to the six faces of the die.

These values are specific and countable, so (Y) is a discrete random variable.

Number of Students in a Class:

Let Z represent the number of students in a classroom. The possible values of Z are 0, 1, 2, 3, ..., as the number of students is a discrete variable and cannot be a fraction (e.g., you can't have 20.5 students).

Since these values are distinct and countable, (Z) is a discrete random variable.

2. Continuous Random Variable

- A continuous random variable can take any value within a given range, with no gaps between the values, forming a continuous range with infinite possibilities.
- It's like a smooth slide, where the variable can assume any point within that range. Unlike discrete variables, which have distinct steps, continuous variables are used for things that can be measured, such as time, height, or temperature.

Examples:

Height, time, and temperature are continuous random variables because they can take an infinite number of values within a range, such as 150 cm, 150.5 cm, 150.55 cm for height, 12.3 seconds, 12.34 seconds for time, and 25.1°C, 25.11°C for temperature. These variables can be measured with.

4. Introduction to Probability Distributions

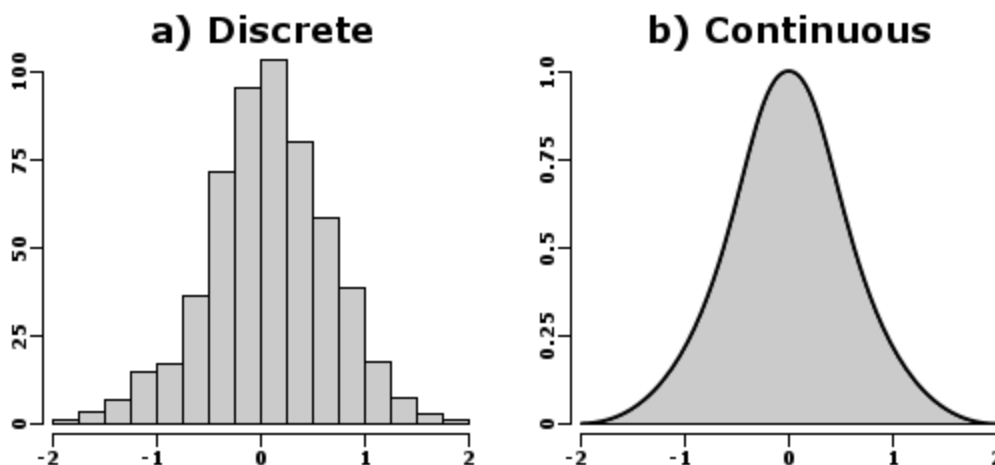
- A probability distribution describes how probabilities are distributed over the possible values of a random variable.

Types of Probability Distributions

Probability distributions can be broadly categorized into two types:

1. Discrete Probability Distributions
2. Continuous Probability Distributions

Let's explore both types with easy examples, manual calculations, and Python code implementation.



Discrete Probability Distributions

- A Discrete Probability Distribution describes probabilities of outcomes for a random event with countable results. Outcomes are distinct, like the number of heads in coin tosses or cars passing a signal.
The sum of probabilities for all outcomes equals 1.

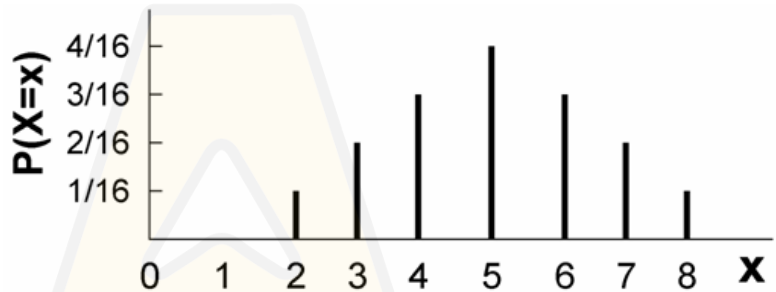
PMF (Probability Mass Function)

PMF calculates probabilities for discrete outcomes.

It assigns a specific probability to each possible value of a discrete random variable.

For a discrete random variable X , $P(X=x)$ gives the probability of X taking value x .

x	$P(x)$
2	$1/16$
3	$2/16$
4	$3/16$
5	$4/16$
6	$3/16$
7	$2/16$
8	$1/16$



How to Calculate the PMF for Discrete Distributions:

1. List all possible values the random variable can take (e.g., heads or tails in a coin toss).
2. Assign a probability to each outcome (e.g., for a fair coin, $P(H) = 0.5$, $P(T) = 0.5$).
3. Ensure the sum of all probabilities equals 1 (e.g., for a fair die, $P(1) = P(2) = \dots = P(6) = 1/6$).
4. **Write the PMF:**

- Express the PMF as a function, showing the probability of each possible outcome. For example, for a fair die:

$$P(X = 1) = \frac{1}{6}, \quad P(X = 2) = \frac{1}{6}, \quad P(X = 3) = \frac{1}{6}, \quad P(X = 4) = \frac{1}{6}, \quad P(X = 5) = \frac{1}{6}, \quad P(X = 6) = \frac{1}{6}$$

The sum of these probabilities equals 1, as expected:

$$P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) = 1$$

Examples of Discrete Random Variables with PMF

Number of Heads in Coin Tosses

When tossing a fair coin twice, the possible outcomes for the number of heads are 0, 1, or 2, depending on how many heads appear in the tosses. The outcomes are:

- 0 heads (both tails)
- 1 head (one head, one tail)
- 2 heads (both heads)

- The probabilities for each outcome:

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{4}$$

(Note: The sum of the probabilities is $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$).

Continuous Probability Distributions

A Continuous Probability Distribution is used when the random variable can take any value within a given range. These values are typically real numbers and cannot be counted individually. Examples include measurements like height, weight, temperature, and time.

Types of Continuous Distributions:

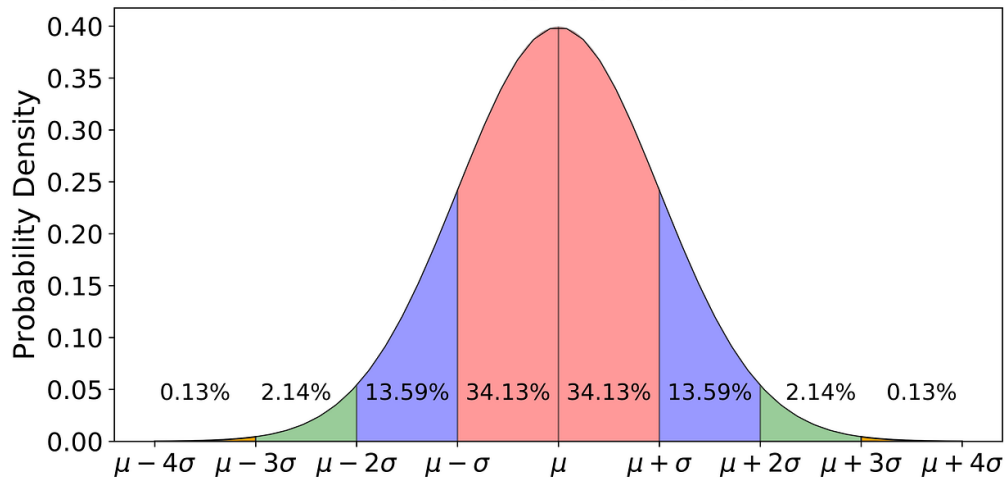
- Normal Distribution**

Let's first look into the **Normal Distribution** as an example.

Normal Distribution (Continuous)

- The Normal Distribution is a bell-shaped curve that is symmetric around the mean. It is commonly used to represent real-world data that tends to cluster around a central value.

Normal Distribution



Problem:

- Find the probability that a randomly selected woman has a height between 62 inches and 68 inches, given that the heights follow a Normal Distribution with a mean of 65 inches and a standard deviation of 3 inches. To calculate this, use the Z-score formula to determine how many standard deviations the values (62 and 68 inches) are from the mean.

The Z-score formula is:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X is the value (height),
- μ is the mean (65 inches),
- σ is the standard deviation (3 inches).

We will calculate the Z-scores for 62 inches and 68 inches.

Step 3: Calculate Z-Scores

1. For 62 inches:

$$Z_{62} = \frac{62 - 65}{3} = \frac{-3}{3} = -1$$

2. For 68 inches:

$$Z_{68} = \frac{68 - 65}{3} = \frac{3}{3} = 1$$

So, the Z-scores for 62 inches and 68 inches are **-1** and **1**, respectively.

Step 4: Use the Standard Normal Distribution

- Find the probability between **Z = -1** and **Z = 1** using a **Z-table** or **Python**.
- Approximately **68% of data** lies within 1 standard deviation of the mean in a normal distribution.

Step 5: Python Code for Normal Distribution Now, let's use Python's `norm.cdf()` function to calculate this probability.

```
In [ ]: from scipy.stats import norm

# Parameters
mu = 65 # Mean height (inches)
sigma = 3 # Standard deviation (inches)

# Z-scores for 62 and 68 inches
z_62 = (62 - mu) / sigma
z_68 = (68 - mu) / sigma

# Probability of being between 62 and 68 inches
probability = norm.cdf(z_68) - norm.cdf(z_62)

print("Probability of height between 62 and 68 inches:", probability)
```

Probability of height between 62 and 68 inches: 0.6826894921370859

Explanation of the Code:

- `from scipy.stats import norm`: Imports the `norm` function to work with normal distributions.
- `mu = 65`: Mean height.
- `sigma = 3`: Standard deviation.
- `norm.cdf(z_68) - norm.cdf(z_62)`: Calculates the probability of a height between 62 and 68 inches.

THANK YOU



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